Results of Experiments

Note: This file contains training results on training data and prediction results on test data concerning the experiments as described in chapter 7 of the master thesis. **In this file the results of the GRU-model are documented only.** Each experiment is conducted two times with different sampled training and validation data as well as with different initial model weights[1]. The sub experiment with different training/validation/initial model weight sets are marked with “random_state” labels 42, 31.

The listed confusion matrices show the ground truth label on the left side, the predicted label on the top.

Experiment	Description
1a	Class 0 = all other, class 1 = SR,
4a-c	Classification results of different label scenarios 4a: class 0 = SR, class 1 = all other, class 2 = binary label, class 3 = single label 4b: class 0 = SR, class 1 = all other, class 2 = form, class 3 = rhythm 4c: class 0 = 'ABQRS', class 1= 'AFIB', class 2 = 'INVT', class 3 = 'LNGQT', class 4 = 'LPR', class 5 = 'LVOLT', class 6 = 'NDT', class 7 = 'NST_' class 8 = 'NT_', class 9 = 'PAC', class 10 = 'PACE', class 11 = 'PVC', class 12 ='QWAVE', class 13 = 'SARRH', class 14 = 'SBRAD', class 15 = 'SR', class 16 = 'STACH', class 17 = 'STD_', class 18 = 'SVARR', class 19 = 'VCLVH', class 20 = 'all_other'
5	Multilabel classification Label: SR, ABQRS, AFIB, all other Resulting eight label combinations: SR, ABQRS, AFIB, all other, ABQRS + all_other, AFIB + all_other, ABQRS + AFIB, ABQRS + AFIB + all_other

[1] According to different starting conditions by different sampled training data and initial model weights of the training the results differ slightly between "random_state" 42, 31 and 7.

Experiment 1a / `Random_state: 42`

```
0='all_other', 1='SR'


Training metrics:
Loss:     0.6174
Accuracy: 0.6723
F1-score: 0.6598
Confusion matrix:
    0    1
0[5099 2445]
1[2266 4568]

Training classification report:
              precision    recall  f1-score       support
class_0        0.692329  0.675901  0.684016   7544.000000
class_1        0.651362  0.668423  0.659782   6834.000000
accuracy       0.672347  0.672347  0.672347      0.672347
macro avg      0.671845  0.672162  0.671899  14378.000000
weighted avg   0.672857  0.672347  0.672497  14378.000000


Test metrics:
Loss:     0.6381
Accuracy: 0.6477
F1-score: 0.6449
Confusion matrix:
   0   1
0[697 416]
1[333 680]

Test classification report:
              precision    recall  f1-score      support
class_0        0.676699  0.626235  0.650490  1113.000000
class_1        0.620438  0.671273  0.644855  1013.000000
accuracy       0.647695  0.647695  0.647695     0.647695
macro avg      0.648568  0.648754  0.647673  2126.000000
weighted avg   0.649892  0.647695  0.647805  2126.000000
```

Experiment 1a / `Random_state: 31`

```
0='all_other', 1='SR'


Training metrics:
Loss:     0.6912
Accuracy: 0.5247
F1-score: 0.1666
Confusion matrix:
    0     1
0[6861  683]
1[6151  683]

Training classification report:
              precision    recall  f1-score      support
class_0        0.527283  0.909464  0.667542   7544.00000
class_1        0.500000  0.099941  0.166585   6834.00000
accuracy       0.524690  0.524690  0.524690      0.52469
macro avg      0.513641  0.504703  0.417064  14378.00000
weighted avg   0.514315  0.524690  0.429433  14378.00000


Test metrics:
Loss:     0.6915
Accuracy: 0.5245
F1-score: 0.1957
Confusion matrix:
   0   1
0[992 121]
1[890 123]

Test classification report:
              precision    recall  f1-score      support
class_0        0.527099  0.891285  0.662437  1113.000000
class_1        0.504098  0.121422  0.195704  1013.000000
accuracy       0.524459  0.524459  0.524459     0.524459
macro avg      0.515599  0.506353  0.429071  2126.000000
weighted avg   0.516140  0.524459  0.440048  2126.000000
```

Experiment 4a / `Random_state: 42`

```
0='SR', 1='all_other', 2='binary_label' 3='single_label'


Training metrics:
Loss:     1.3215
Accuracy: 0.4711
F1-score: 0.2166
Confusion matrix:
     0    1    2    3
0[5954  367   94  419]
1[ 652   77    9   38]
2[ 869   48   11   48]
3[4262  259   52  298]

Training classification report:
              precision    recall  f1-score      support
class_0        0.507285  0.871232  0.641215   6834.00000
class_1        0.102530  0.099227  0.100851    776.00000
class_2        0.066265  0.011270  0.019264    976.00000
class_3        0.371108  0.061178  0.105041   4871.00000
accuracy       0.471130  0.471130  0.471130      0.47113
macro avg      0.261797  0.260727  0.216593  13457.00000
weighted avg   0.402667  0.471130  0.370868  13457.00000


Test metrics:
Loss:     1.3280
Accuracy: 0.4594
F1-score: 0.2152
Confusion matrix:
    0   1   2   3
0[856  77  10  70]
1[ 93  13   0   7]
2[125  10   2   8]
3[619  52   7  45]

Test classification report:
              precision    recall  f1-score      support
class_0        0.505611  0.845015  0.632668  1013.000000
class_1        0.085526  0.115044  0.098113   113.000000
class_2        0.105263  0.013793  0.024390   145.000000
class_3        0.346154  0.062241  0.105510   723.000000
accuracy       0.459378  0.459378  0.459378     0.459378
macro avg      0.260639  0.259023  0.215170  1994.000000
weighted avg   0.394875  0.459378  0.367001  1994.000000
```

Experiment 4a / `Random_state: 31`

```
0='SR', 1='all_other', 2='binary_label' 3='single_label'


Training metrics:
Loss:     1.0681
Accuracy: 0.5147
F1-score: 0.3610
Confusion matrix:
     0    1    2    3
0[5520  263  567  484]
0[ 172  250  221  133]
2[ 292  181  320  183]
3[2579  415 1041  836]

Training classification report:
              precision    recall  f1-score       support
class_0        0.644634  0.807726  0.717023   6834.000000
class_1        0.225428  0.322165  0.265252    776.000000
class_2        0.148906  0.327869  0.204800    976.000000
class_3        0.511002  0.171628  0.256954   4871.000000
accuracy       0.514676  0.514676  0.514676      0.514676
macro avg      0.382493  0.407347  0.361007  13457.000000
weighted avg   0.536136  0.514676  0.487291  13457.000000


Test metrics:
Loss:     1.0854
Accuracy: 0.4930
F1-score: 0.3243
Confusion matrix:
    0   1   2   3
0[812  55  73  73]
1[ 27  28  34  24]
2[ 47  22  39  37]
3[417  67 135 104]

Test classification report:
              precision    recall  f1-score      support
class_0        0.623177  0.801579  0.701209  1013.000000
class_1        0.162791  0.247788  0.196491   113.000000
class_2        0.138790  0.268966  0.183099   145.000000
class_3        0.436975  0.143845  0.216441   723.000000
accuracy       0.492979  0.492979  0.492979     0.492979
macro avg      0.340433  0.365544  0.324310  1994.000000
weighted avg   0.494349  0.492979  0.459160  1994.000000
```

Experiment 4b / `Random_state: 42`

```
0='SR', 1='all_other', 2='form', 3='rhythm'


Training metrics:
Loss:     1.3634
Accuracy: 0.4123
F1-score: 0.2261
Confusion matrix:

     0    1    2    3
0[5376  869   99  490]
1[1990  351   45  183]
2[2901  390   40  197]
3[1092  170   24  162]

Training classification report:
              precision    recall  f1-score       support
class_0        0.473281  0.786655  0.590997   6834.000000
class_1        0.197191  0.136629  0.161416   2569.000000
class_2        0.192308  0.011338  0.021413   3528.000000
class_3        0.156977  0.111878  0.130645   1448.000000
accuracy       0.412337  0.412337  0.412337      0.412337
macro avg      0.254939  0.261625  0.226118  14379.000000
weighted avg   0.323162  0.412337  0.328136  14379.000000


Test metrics:
Loss:     1.3653
Accuracy: 0.4045
F1-score: 0.2283
Confusion matrix:
    0   1   2   3
0[769 141  18  85]
1[275  57   8  28]
2[414  58   9  36]
3[166  34   3  25]

Test classification report:
              precision    recall  f1-score      support
class_0        0.473522  0.759131  0.583239  1013.000000
class_1        0.196552  0.154891  0.173252   368.000000
class_2        0.236842  0.017408  0.032432   517.000000
class_3        0.143678  0.109649  0.124378   228.000000
accuracy       0.404516  0.404516  0.404516     0.404516
macro avg      0.262649  0.260270  0.228325  2126.000000
weighted avg   0.332651  0.404516  0.329117  2126.000000
```

Experiment 4b / `Random_state: 31`

```
0='SR', 1='all_other', 2='form', 3='rhythm'


Training metrics:
Loss:     1.0755
Accuracy: 0.5714
F1-score: 0.4944
Confusion matrix:
     0    1    2    3
0[4921  381 1343  189]
1[ 347 1382  646  194]
2[1319  566 1538  105]
3[ 433  420  220  375]

Training classification report:
              precision    recall  f1-score       support
class_0        0.700997  0.720076  0.710409   6834.000000
class_1        0.502728  0.537953  0.519744   2569.000000
class_2        0.410462  0.435941  0.422818   3528.000000
class_3        0.434531  0.258978  0.324535   1448.000000
accuracy       0.571389  0.571389  0.571389      0.571389
macro avg      0.512179  0.488237  0.494376  14379.000000
weighted avg   0.567455  0.571389  0.566923  14379.000000


Test metrics:
Loss:     1.1375
Accuracy: 0.5306
F1-score: 0.4460
Confusion matrix:
    0   1   2   3
0[709  68 202  34]
1[ 60 186 101  21]
2[229  87 188  13]
3[ 80  58  45  45]

Test classification report:
              precision    recall  f1-score      support
class_0        0.657699  0.699901  0.678144  1013.000000
class_1        0.466165  0.505435  0.485007   368.000000
class_2        0.350746  0.363636  0.357075   517.000000
class_3        0.398230  0.197368  0.263930   228.000000
accuracy       0.530574  0.530574  0.530574     0.530574
macro avg      0.468210  0.441585  0.446039  2126.000000
weighted avg   0.522075  0.530574  0.522214  2126.000000
```

Experiment 4c / `Random_state: 42`

0='ABQRS', 1='AFIB', 2='INVT', 3='LNGQT', 4='LPR', 5='LVOLT', 6='NDT', 7='NST_' 8='NT_', 9='PAC', 10='PACE',11='PVC', 12='QWAVE', 13='SARRH', 14='SBRAD', **15='SR'**, 16='STACH', 17='STD_', 18='SVARR', 19='VCLVH', **20='all_other'**

Training metrics:
Loss: 3.0392
Accuracy: 0.0138
F1-score: 0.0114
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	0	0	6	0	2	139	0	8	1	461	148	31	103	2	0	7	122	1	369	185	0]
1 [	0	0	3	0	0	26	0	3	0	65	41	4	27	0	0	0	32	1	79	41	0]
2 [	0	0	0	0	0	4	0	0	0	4	6	0	3	0	0	0	4	0	9	6	0]
3 [	0	0	0	0	0	8	0	0	0	10	2	2	3	0	0	1	3	0	7	7	0]
4 [	0	0	0	0	2	14	0	1	0	13	13	1	10	0	0	1	11	0	15	16	0]
5 [	0	0	0	0	0	5	0	0	0	20	4	2	4	0	0	0	3	0	11	10	0]
6 [	0	0	2	0	1	59	0	2	0	160	57	10	38	2	0	0	50	0	115	75	0]
7 [	0	0	1	0	3	24	0	2	0	41	25	5	14	1	0	2	10	0	35	20	0]
8 [	0	0	0	0	0	6	0	1	0	14	10	6	9	0	0	0	5	0	14	5	0]
9 [	0	0	1	0	0	10	0	1	0	13	13	5	14	0	0	0	9	0	13	6	0]
10[	0	0	0	0	1	5	0	0	0	24	48	3	23	0	0	0	51	0	23	13	0]
11[	0	0	0	0	0	19	0	1	0	36	43	5	28	0	0	1	25	0	33	20	0]
12[	0	0	0	0	0	12	0	2	0	30	21	4	18	0	0	2	15	0	36	12	0]
13[	0	0	1	0	3	34	0	1	1	61	53	9	25	0	0	1	47	0	47	34	0]
14[	0	0	1	0	0	30	0	1	0	63	30	6	26	0	0	0	29	0	55	47	0]
15[	**0**	**0**	**36**	**0**	**21**	**681**	**1**	**87**	**5**	**1492**	**948**	**181**	**586**	**2**	**0**	**16**	**822**	**0**	**1201**	**755**	**0]**
16[	0	0	1	0	1	18	0	1	1	47	45	8	24	0	0	0	40	0	45	15	0]
17[	0	0	0	0	0	9	0	0	2	22	13	3	11	0	0	0	18	0	7	11	0]
18[	0	0	0	0	0	1	0	0	0	12	1	0	4	0	0	0	6	0	17	9	0]
19[	0	0	1	0	0	35	0	3	0	60	34	9	27	0	0	1	25	0	42	33	0]
20[	**0**	**0**	**12**	**0**	**6**	**210**	**0**	**41**	**1**	**586**	**412**	**81**	**222**	**3**	**0**	**9**	**292**	**0**	**542**	**256**	**0]**

Training classification report:

	precision	recall	f1-score	support
class_0	0.000000	0.000000	0.000000	1585.00000
class_1	0.000000	0.000000	0.000000	322.00000
class_2	0.000000	0.000000	0.000000	36.00000
class_3	0.000000	0.000000	0.000000	43.00000
class_4	0.050000	0.020619	0.029197	97.00000
class_5	0.003706	0.084746	0.007102	59.00000
class_6	0.000000	0.000000	0.000000	571.00000
class_7	0.012903	0.010929	0.011834	183.00000
class_8	0.000000	0.000000	0.000000	70.00000
class_9	0.004020	0.152941	0.007834	85.00000
class_10	0.024403	0.251309	0.044486	191.00000
class_11	0.013333	0.023697	0.017065	211.00000
class_12	0.014766	0.118421	0.026258	152.00000
class_13	0.000000	0.000000	0.000000	317.00000
class_14	0.000000	0.000000	0.000000	288.00000
class_15	0.390244	0.002341	0.004655	6834.00000
class_16	0.024707	0.162602	0.042895	246.00000
class_17	0.000000	0.000000	0.000000	96.00000
class_18	0.006262	0.340000	0.012297	50.00000
class_19	0.020939	0.122222	0.035753	270.00000
class_20	0.000000	0.000000	0.000000	2673.00000
accuracy	0.013840	0.013840	0.013840	0.01384
macro avg	0.026918	0.061420	0.011399	14379.00000
weighted avg	0.187528	0.013840	0.005202	14379.00000

Test metrics:
Loss: 3.0379
Accuracy: 0.0132
F1-score: 0.0095
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0	0	0	0	0	0	16	0	4	0	60	32	4	19	1	0	0	17	0	41	25	0
1	0	0	1	0	0	4	0	1	1	13	8	3	6	0	0	0	4	0	8	2	0
2	0	0	0	0	0	2	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	0	2	0	0
4	0	0	0	0	0	1	0	0	0	2	0	1	2	0	0	0	0	0	1	0	0
5	0	0	0	0	0	1	0	0	0	2	0	1	0	0	0	0	1	0	4	0	0
6	0	0	0	0	0	9	0	1	0	20	18	2	5	0	0	0	10	0	27	9	0
7	0	0	0	0	0	4	0	0	0	7	6	0	1	0	0	0	4	0	4	1	0
8	0	0	0	0	0	1	0	0	0	1	1	1	2	0	0	0	0	0	2	0	0
9	0	0	0	0	0	2	0	0	0	4	1	1	1	0	0	0	2	0	0	3	0
10	0	0	0	0	1	0	0	0	0	3	10	0	4	0	0	0	6	0	2	0	0
11	0	0	0	0	0	3	0	0	0	1	13	2	4	0	0	0	7	0	4	1	0
12	0	0	0	0	0	1	0	0	0	4	6	3	0	0	0	0	2	0	3	3	0
13	0	0	0	0	1	10	0	0	0	8	11	0	6	0	0	0	3	0	5	5	0
14	0	0	0	0	0	3	0	0	0	9	8	1	4	0	0	0	1	0	12	6	0
15	**0**	**0**	**5**	**0**	**1**	**127**	**0**	**20**	**0**	**177**	**174**	**26**	**96**	**2**	**0**	**0**	**130**	**0**	**162**	**93**	**0**
16	0	0	0	0	1	2	0	0	0	8	6	1	3	0	0	0	7	0	11	3	0
17	0	0	0	0	0	2	0	0	0	1	2	0	4	0	0	0	0	0	3	2	0
18	0	0	0	0	0	1	0	0	0	4	0	0	0	0	0	0	1	0	1	0	0
19	0	0	0	0	0	7	0	0	0	8	7	2	1	1	0	0	2	0	8	3	0
20	**0**	**0**	**0**	**0**	**0**	**35**	**0**	**3**	**1**	**79**	**76**	**13**	**32**	**1**	**0**	**1**	**45**	**0**	**83**	**21**	**0**

Test classification report:

	precision	recall	f1-score	support
class_0	0.000000	0.000000	0.000000	219.00000
class_1	0.000000	0.000000	0.000000	51.00000
class_2	0.000000	0.000000	0.000000	4.00000
class_3	0.000000	0.000000	0.000000	5.00000
class_4	0.000000	0.000000	0.000000	7.00000
class_5	0.004329	0.111111	0.008333	9.00000
class_6	0.000000	0.000000	0.000000	101.00000
class_7	0.000000	0.000000	0.000000	27.00000
class_8	0.000000	0.000000	0.000000	8.00000
class_9	0.009662	0.285714	0.018692	14.00000
class_10	0.026316	0.384615	0.049261	26.00000
class_11	0.032787	0.057143	0.041667	35.00000
class_12	0.000000	0.000000	0.000000	22.00000
class_13	0.000000	0.000000	0.000000	49.00000
class_14	0.000000	0.000000	0.000000	44.00000
class_15	0.000000	0.000000	0.000000	1013.00000
class_16	0.028807	0.166667	0.049123	42.00000
class_17	0.000000	0.000000	0.000000	14.00000
class_18	0.002611	0.142857	0.005128	7.00000
class_19	0.016949	0.076923	0.027778	39.00000
class_20	0.000000	0.000000	0.000000	390.00000
accuracy	0.013170	0.013170	0.013170	0.01317
macro avg	0.005784	0.058335	0.009523	2126.00000
weighted avg	0.001832	0.013170	0.002944	2126.00000

Experiment 4c / Random_state: 42 / "learning_rate": 0.5e-3,

0='ABQRS', 1='AFIB', 2='INVT', 3='LNGQT', 4='LPR', 5='LVOLT', 6='NDT', 7='NST_' 8='NT_', 9='PAC', 10='PACE',11='PVC', 12='QWAVE', 13='SARRH', 14='SBRAD', **15='SR'**, 16='STACH', 17='STD_', 18='SVARR', 19='VCLVH', **20='all_other'**

Training metrics:
Loss: 3.0297
Accuracy: 0.0180
F1-score: 0.0131
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	5	0	17	0	15	0	96	2	1	0	275	80	1	205	0	0	207	0	575	95	11]
1 [	0	0	3	0	6	0	24	0	0	0	72	15	0	42	0	0	38	0	98	22	2]
2 [	0	0	0	0	1	0	3	0	0	0	10	1	0	1	0	0	5	0	11	4	0]
3 [	0	0	1	0	0	0	2	0	0	0	10	2	0	7	0	0	5	0	10	6	0]
4 [	0	0	1	0	3	0	11	0	0	0	31	6	0	7	0	0	15	0	17	6	0]
5 [	0	0	1	0	0	0	1	0	0	0	11	3	0	8	0	0	9	0	23	3	0]
6 [	0	0	8	0	7	0	39	0	0	0	99	29	1	76	0	0	72	0	193	39	8]
7 [	0	0	3	0	6	0	17	2	0	0	48	9	0	25	0	0	13	0	47	11	2]
8 [	0	0	2	0	1	0	3	1	0	0	17	4	0	10	0	0	8	0	22	2	0]
9 [	0	0	2	0	3	0	6	0	0	0	21	8	0	16	0	0	12	0	14	2	1]
10[	0	0	2	0	10	0	9	1	0	0	72	11	0	12	0	0	34	0	32	4	4]
11[	0	0	5	0	8	0	8	2	0	0	73	15	0	27	0	0	23	0	37	11	2]
12[	0	0	2	0	3	0	6	0	0	0	33	12	0	19	0	0	30	0	37	10	0]
13[	2	0	4	0	4	0	32	2	0	0	83	22	0	40	0	0	44	0	67	12	5]
14[	0	0	3	0	1	0	34	0	0	0	54	21	0	45	0	0	26	0	87	16	1]
15[	**8**	**0**	**112**	**0**	**116**	**0**	**474**	**27**	**0**	**0**	**1615**	**476**	**7**	**904**	**0**	**0**	**898**	**0**	**1782**	**368**	**47]**
16[	0	0	4	0	6	0	10	1	0	0	69	23	0	25	0	0	43	0	58	7	0]
17[	0	0	3	0	2	0	8	0	0	0	25	9	0	8	0	0	20	0	17	3	1]
18[	0	0	0	0	0	0	2	0	0	0	5	1	0	7	0	0	7	0	21	7	0]
19[	1	0	4	0	4	0	24	0	0	0	75	16	0	26	0	0	21	0	85	13	1]
20[	**5**	**0**	**49**	**0**	**64**	**0**	**175**	**8**	**0**	**0**	**622**	**197**	**1**	**345**	**0**	**0**	**335**	**0**	**757**	**109**	**6]**

Training classification report:

	precision	recall	f1-score	support
class_0	0.238095	0.003155	0.006227	1585.000000
class_1	0.000000	0.000000	0.000000	322.000000
class_2	0.000000	0.000000	0.000000	36.000000
class_3	0.000000	0.000000	0.000000	43.000000
class_4	0.011538	0.030928	0.016807	97.000000
class_5	0.000000	0.000000	0.000000	59.000000
class_6	0.039634	0.068301	0.050161	571.000000
class_7	0.043478	0.010929	0.017467	183.000000
class_8	0.000000	0.000000	0.000000	70.000000
class_9	0.000000	0.000000	0.000000	85.000000
class_10	0.021687	0.376963	0.041014	191.000000
class_11	0.015625	0.071090	0.025619	211.000000
class_12	0.000000	0.000000	0.000000	152.000000
class_13	0.021563	0.126183	0.036832	317.000000
class_14	0.000000	0.000000	0.000000	288.000000
class_15	0.000000	0.000000	0.000000	6834.000000
class_16	0.023056	0.174797	0.040739	246.000000
class_17	0.000000	0.000000	0.000000	96.000000
class_18	0.005263	0.420000	0.010396	50.000000
class_19	0.017333	0.048148	0.025490	270.000000
class_20	0.065934	0.002245	0.004342	2673.000000
accuracy	0.018012	0.018012	0.018012	0.018012
macro avg	0.023962	0.063464	0.013100	14379.000000
weighted avg	0.042438	0.018012	0.006766	14379.000000

Test metrics:
Loss: 3.0291
Accuracy: 0.0221
F1-score: 0.0144
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	0	0	1	0	4	0	20	1	0	0	46	25	0	21	0	0	27	0	66	7	1]
1 [	1	0	1	0	3	0	4	1	0	0	11	1	0	6	0	0	6	0	14	3	0]
2 [	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	1	1	0]
3 [	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1	0	2	0	0]
4 [	0	0	0	0	0	0	1	0	0	0	1	1	0	3	0	0	1	0	0	0	0]
5 [	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	2	0	3	0	0]
6 [	1	0	2	0	3	0	9	0	0	0	20	6	0	12	0	0	14	0	29	5	0]
7 [	0	0	0	0	0	0	2	0	0	0	7	2	0	9	0	0	2	0	4	0	1]
8 [	0	0	0	0	0	0	0	0	0	0	3	0	0	2	0	0	0	0	3	0	0]
9 [	0	0	0	0	0	0	0	0	0	0	5	1	0	3	0	0	2	0	3	0	0]
10[	1	0	1	0	4	0	0	0	0	0	10	2	0	0	0	0	4	0	3	0	1]
11[	0	0	1	0	1	0	1	0	0	0	15	7	0	1	0	0	5	0	4	0	0]
12[	0	0	1	0	2	0	5	0	0	0	1	1	0	2	0	0	1	0	7	2	0]
13[	0	0	0	0	2	0	5	0	0	0	19	2	0	6	0	0	2	0	12	1	0]
14[	1	0	1	0	0	0	3	0	0	0	9	4	0	6	0	0	6	0	12	2	0]
15[	**1**	**0**	**22**	**0**	**20**	**0**	**66**	**8**	**0**	**0**	**289**	**77**	**0**	**110**	**0**	**0**	**141**	**0**	**213**	**52**	**14]**
16[	0	0	1	0	0	0	2	0	0	0	12	2	0	4	0	0	7	0	12	2	0]
17[	0	0	0	0	0	0	2	0	0	0	6	0	0	1	0	0	2	0	2	1	0]
18[	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	1	1]
19[	0	0	0	0	0	0	3	0	0	0	8	4	0	10	0	0	6	0	7	1	0]
20[	**0**	**0**	**4**	**0**	**9**	**0**	**23**	**1**	**0**	**0**	**102**	**35**	**1**	**46**	**0**	**0**	**54**	**0**	**98**	**15**	**2]**

Test classification report:

	precision	recall	f1-score	support
class_0	0.000000	0.000000	0.000000	219.000000
class_1	0.000000	0.000000	0.000000	51.000000
class_2	0.000000	0.000000	0.000000	4.000000
class_3	0.000000	0.000000	0.000000	5.000000
class_4	0.000000	0.000000	0.000000	7.000000
class_5	0.000000	0.000000	0.000000	9.000000
class_6	0.061644	0.089109	0.072874	101.000000
class_7	0.000000	0.000000	0.000000	27.000000
class_8	0.000000	0.000000	0.000000	8.000000
class_9	0.000000	0.000000	0.000000	14.000000
class_10	0.017699	0.384615	0.033841	26.000000
class_11	0.041176	0.200000	0.068293	35.000000
class_12	0.000000	0.000000	0.000000	22.000000
class_13	0.024194	0.122449	0.040404	49.000000
class_14	0.000000	0.000000	0.000000	44.000000
class_15	0.000000	0.000000	0.000000	1013.000000
class_16	0.024648	0.166667	0.042945	42.000000
class_17	0.000000	0.000000	0.000000	14.000000
class_18	0.010000	0.714286	0.019724	7.000000
class_19	0.010753	0.025641	0.015152	39.000000
class_20	0.100000	0.005128	0.009756	390.000000
accuracy	0.022107	0.022107	0.022107	0.022107
macro avg	0.013815	0.081328	0.014428	2126.000000
weighted avg	0.023442	0.022107	0.008912	2126.000000

Experiment 4c / `Random_state: 31` / "learning_rate": 0.5e-3,

0='ABQRS', 1='AFIB', 2='INVT', 3='LNGQT', 4='LPR', 5='LVOLT', 6='NDT', 7='NST_' 8='NT_', 9='PAC', 10='PACE',11='PVC', 12='QWAVE', 13='SARRH', 14='SBRAD', **15='SR'**, 16='STACH', 17='STD_', 18='SVARR', 19='VCLVH', **20='all_other'**

Training metrics:
Loss: 3.0234
Accuracy: 0.0208
F1-score: 0.0209
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	7	0	546	5	0	62	69	3	41	2	132	27	436	25	120	0	16	0	74	18	2]
1 [	0	0	113	1	0	17	9	0	8	0	45	7	79	5	15	0	4	0	14	5	0]
2 [	1	0	15	0	0	2	0	0	1	0	4	1	10	0	1	0	0	0	1	0	0]
3 [	0	0	17	0	0	1	3	0	0	0	4	1	5	1	4	0	2	0	5	0	0]
4 [	0	0	39	0	0	6	2	0	5	0	7	2	25	0	7	0	1	0	3	0	0]
5 [	0	0	30	1	0	4	1	0	1	0	0	1	15	1	5	0	0	0	0	0	0]
6 [	1	0	231	1	0	27	24	1	12	2	47	7	137	9	43	0	4	0	24	1	0]
7 [	0	0	57	2	0	13	3	1	7	0	19	9	39	5	12	0	3	0	10	3	0]
8 [	0	0	23	0	0	4	1	0	2	0	3	2	23	2	5	0	0	0	4	1	0]
9 [	0	0	24	0	1	9	1	0	3	1	11	3	18	1	6	0	0	0	6	1	0]
10[	0	0	19	0	0	2	3	1	3	0	131	0	19	3	4	0	0	0	5	1	0]
11[	0	0	63	0	1	10	7	0	5	2	47	14	40	1	8	1	3	0	6	2	1]
12[	1	0	47	1	0	7	3	0	3	1	12	7	41	8	13	0	1	0	5	2	0]
13[	0	0	102	2	0	17	11	0	5	0	57	10	56	14	13	0	2	0	23	3	2]
14[	2	0	127	1	1	14	6	0	5	0	23	8	51	8	22	0	1	0	15	4	0]
15[	**17**	**0**	**2407**	**37**	**8**	**347**	**235**	**21**	**181**	**13**	**818**	**197**	**1355**	**199**	**422**	**4**	**88**	**1**	**371**	**94**	**19]**
16[	1	0	54	0	0	16	8	0	9	0	53	9	50	8	13	0	6	0	11	8	0]
17[	0	0	28	0	0	6	3	0	3	0	16	3	23	2	8	0	0	0	4	0	0]
18[	0	0	16	0	0	0	0	0	2	0	11	1	10	0	6	0	0	0	4	0	0]
19[	0	0	91	0	0	17	12	0	6	0	41	9	43	10	17	0	5	0	16	2	1]
20[	**9**	**0**	**833**	**11**	**4**	**143**	**82**	**8**	**74**	**9**	**383**	**73**	**646**	**59**	**162**	**1**	**21**	**1**	**121**	**26**	**7]**

Training classification report:

	precision	recall	f1-score	support
class_0	0.179487	0.004416	0.008621	1585.000000
class_1	0.000000	0.000000	0.000000	322.000000
class_2	0.003073	0.416667	0.006100	36.000000
class_3	0.000000	0.000000	0.000000	43.000000
class_4	0.000000	0.000000	0.000000	97.000000
class_5	0.005525	0.067797	0.010217	59.000000
class_6	0.049689	0.042032	0.045541	571.000000
class_7	0.028571	0.005464	0.009174	183.000000
class_8	0.005319	0.028571	0.008969	70.000000
class_9	0.033333	0.011765	0.017391	85.000000
class_10	0.070279	0.685864	0.127494	191.000000
class_11	0.035806	0.066351	0.046512	211.000000
class_12	0.013137	0.269737	0.025053	152.000000
class_13	0.038781	0.044164	0.041298	317.000000
class_14	0.024283	0.076389	0.036851	288.000000
class_15	0.666667	0.000585	0.001170	6834.000000
class_16	0.038217	0.024390	0.029777	246.000000
class_17	0.000000	0.000000	0.000000	96.000000
class_18	0.005540	0.080000	0.010363	50.000000
class_19	0.011696	0.007407	0.009070	270.000000
class_20	0.218750	0.002619	0.005176	2673.000000
accuracy	0.020794	0.020794	0.020794	0.020794
macro avg	0.068007	0.087344	0.020894	14379.000000
weighted avg	0.383723	0.020794	0.009602	14379.000000

Test metrics:
Loss: 3.0269
Accuracy: 0.0193
F1-score: 0.0183
Confusion matrix:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
0 [	0	0	75	1	0	17	9	0	2	0	23	8	55	2	18	0	0	0	3	6	0]
1 [	0	0	22	0	0	4	1	0	1	0	7	2	6	3	2	0	0	0	3	0	0]
2 [	0	0	2	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0]
3 [	0	0	0	0	0	0	0	0	0	0	1	0	3	0	0	0	0	0	1	0	0]
4 [	0	0	1	0	0	1	0	0	0	0	2	0	1	1	1	0	0	0	0	0	0]
5 [	0	0	1	0	0	0	0	0	0	0	0	0	6	1	0	0	1	0	0	0	0]
6 [	0	0	32	0	0	7	7	0	2	0	4	3	33	4	4	0	0	0	4	1	0]
7 [	0	0	11	0	0	1	2	0	1	0	4	0	4	2	1	0	0	0	1	0	0]
8 [	0	0	0	0	0	0	0	0	2	0	0	1	2	1	1	0	0	0	0	1	0]
9 [	0	0	4	0	0	0	0	0	0	0	4	0	0	1	4	0	1	0	0	0	0]
10[	0	0	2	0	0	1	0	0	0	0	20	0	2	0	1	0	0	0	0	0	0]
11[	0	0	5	0	0	3	0	0	2	0	12	2	5	2	2	0	0	0	2	0	0]
12[	0	0	13	0	0	1	0	0	0	0	1	1	4	0	0	0	0	0	1	0	1]
13[	0	0	22	0	0	1	1	0	2	0	8	2	7	0	4	0	0	0	0	2	0]
14[	0	0	14	0	0	2	2	0	2	0	2	2	15	1	2	0	1	0	1	0	0]
15[	**2**	**0**	**344**	**6**	**0**	**67**	**29**	**1**	**24**	**2**	**134**	**28**	**187**	**38**	**57**	**0**	**23**	**1**	**51**	**12**	**7]**
16[	0	0	10	0	0	1	1	0	3	0	7	2	10	1	2	0	1	0	4	0	0]
17[	0	0	4	0	0	0	0	0	0	0	2	2	5	0	0	0	1	0	0	0	0]
18[	0	0	3	0	0	0	0	0	0	0	1	0	2	0	0	0	1	0	0	0	0]
19[	0	0	14	0	0	2	0	0	3	0	4	0	11	1	0	0	0	0	4	0	0]
20[	**1**	**0**	**117**	**0**	**0**	**10**	**12**	**0**	**16**	**1**	**61**	**11**	**88**	**8**	**35**	**0**	**6**	**1**	**19**	**3**	**1]**

Test classification report:

	precision	recall	f1-score	support
class_0	0.000000	0.000000	0.000000	219.000000
class_1	0.000000	0.000000	0.000000	51.000000
class_2	0.002874	0.500000	0.005714	4.000000
class_3	0.000000	0.000000	0.000000	5.000000
class_4	0.000000	0.000000	0.000000	7.000000
class_5	0.000000	0.000000	0.000000	9.000000
class_6	0.109375	0.069307	0.084848	101.000000
class_7	0.000000	0.000000	0.000000	27.000000
class_8	0.033333	0.250000	0.058824	8.000000
class_9	0.000000	0.000000	0.000000	14.000000
class_10	0.067340	0.769231	0.123839	26.000000
class_11	0.031250	0.057143	0.040404	35.000000
class_12	0.008949	0.181818	0.017058	22.000000
class_13	0.000000	0.000000	0.000000	49.000000
class_14	0.014815	0.045455	0.022346	44.000000
class_15	0.000000	0.000000	0.000000	1013.000000
class_16	0.028571	0.023810	0.025974	42.000000
class_17	0.000000	0.000000	0.000000	14.000000
class_18	0.000000	0.000000	0.000000	7.000000
class_19	0.000000	0.000000	0.000000	39.000000
class_20	0.111111	0.002564	0.005013	390.000000
accuracy	0.019285	0.019285	0.019285	0.019285
macro avg	0.019410	0.090444	0.018287	2126.000000
weighted avg	0.028011	0.019285	0.008514	2126.000000

Experiment 5 / `Random_state: 42`

```
Training metrics:
Loss:                0.7319
Binary accuracy:     0.6877
Exact-match accuracy: 0.3178
Macro F1-score:      0.5217

Training label-wise confusion matrices:

SR:
      TN   TP
TN [4464 3080]
TP [2095 4739]

ABQRS:
      TN   TP
TN [6835 5289]
TP [ 522 1732]

AFIB:
       TN    TP
TN [12002  1355]
TP [   45   976]

all_other:
      TN   TP
TN [6191 2692]
TP [2881 2614]

0='SR', 1='ABQRS', 2='AFIB', 3='all_other_alone', 4='ABQRS + all_other', 5='AFIB +
all_other', 6='ABQRS + AFIB', 7='ABQRS + AFIB + all_other' 8='invalid_combination'

Training combination-class confusion matrix:
     0    1    2    3    4    5    6    7    8
0[3625  812    0  168  715   98    0  286 1130]
1[ 327  450    0   26  369   16    0  156  241]
2[   3    0    1    5    7   45    0  261    0]
3[1673  545    0  213  638  184    0  502  714]
4[  79  121    0    8   96   11    0  102   52]
5[   3    4    0    2   12  135    0  343    0]
6[   0    1    0    0    3    8    0  129    1]
7[   1    1    0    0    2    4    0   50    0]
8[   0    0    0    0    0    0    0    0    0]

Training multi-label classification report:
              precision    recall  f1-score  support
SR             0.606088  0.693445  0.646830   6834.0
ABQRS          0.246689  0.768412  0.373477   2254.0
AFIB           0.418704  0.955926  0.582339   1021.0
all_other      0.492650  0.475705  0.484029   5495.0
micro avg      0.447613  0.644771  0.528400  15604.0
macro avg      0.441033  0.723372  0.521669  15604.0
weighted avg   0.501964  0.644771  0.545794  15604.0
samples avg    0.477709  0.628588  0.523680  15604.0
```

Training combination-class classification report:

	precision	recall	f1-score	support
SR	0.634740	0.530436	0.577919	6834.000000
ABQRS	0.232678	0.283912	0.255754	1585.000000
AFIB	1.000000	0.003106	0.006192	322.000000
all_other_alone	0.504739	0.047662	0.087099	4469.000000
ABQRS + all_other	0.052117	0.204691	0.083081	469.000000
AFIB + all_other	0.269461	0.270541	0.270000	499.000000
ABQRS + AFIB	0.000000	0.000000	0.000000	142.000000
ABQRS + AFIB + all_other	0.027337	0.862069	0.052994	58.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.317847	0.317847	0.317847	0.317847
macro avg	0.302341	0.244713	0.148116	14378.000000
weighted avg	0.517790	0.317847	0.342390	14378.000000

Test metrics:

Loss:	0.7917
Binary accuracy:	0.6714
Exact-match accuracy:	0.3015
Macro F1-score:	0.4866

Test label-wise confusion matrices:

SR:

	TN	TP
TN	620	493
TP	335	678

ABQRS:

	TN	TP
TN	1072	732
TP	118	204

AFIB:

	TN	TP
TN	1758	216
TP	15	137

all_other:

	TN	TP
TN	866	451
TP	434	375

Test combination-class confusion matrix:

	0	1	2	3	4	5	6	7	8
0	525	105	0	47	99	22	0	55	160
1	67	41	0	11	36	3	0	25	36
2	1	1	0	0	1	10	0	38	0
3	260	88	0	41	87	30	0	68	110
4	13	11	0	2	12	0	0	13	7
5	0	0	0	2	2	15	0	35	2
6	0	0	0	1	3	4	0	26	0
7	0	0	0	0	2	2	0	7	0
8	0	0	0	0	0	0	0	0	0

Test multi-label classification report:

	precision	recall	f1-score	support
SR	0.578992	0.669299	0.620879	1013.0
ABQRS	0.217949	0.633540	0.324324	322.0
AFIB	0.388102	0.901316	0.542574	152.0
all_other	0.453995	0.463535	0.458716	809.0
micro avg	0.424224	0.607143	0.499463	2296.0
macro avg	0.409760	0.666923	0.486623	2296.0
weighted avg	0.471678	0.607143	0.516967	2296.0
samples avg	0.450376	0.591330	0.493321	2296.0

Test combination-class classification report:

	precision	recall	f1-score	support
SR	0.606236	0.518263	0.558808	1013.000000
ABQRS	0.166667	0.187215	0.176344	219.000000
AFIB	0.000000	0.000000	0.000000	51.000000
all_other_alone	0.394231	0.059942	0.104061	684.000000
ABQRS + all_other	0.049587	0.206897	0.080000	58.000000
AFIB + all_other	0.174419	0.267857	0.211268	56.000000
ABQRS + AFIB	0.000000	0.000000	0.000000	34.000000
ABQRS + AFIB + all_other	0.026217	0.636364	0.050360	11.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.301505	0.301505	0.301505	0.301505
macro avg	0.157484	0.208504	0.131204	2126.000000
weighted avg	0.438947	0.301505	0.325915	2126.000000

Experiment 5 / `Random_state: 31`

```
Training metrics:
Loss:                 0.7056
Binary accuracy:      0.7123
Exact-match accuracy: 0.3373
Macro F1-score:       0.5452

Training label-wise confusion matrices:

SR:
      TN   TP
TN [3719 3825]
TP [1249 5585]

ABQRS:
      TN   TP
TN [7336 4788]
TP [ 678 1576]

AFIB:
       TN    TP
TN [12311  1046]
TP [   44   977]

all_other:
      TN   TP
TN [7140 1743]
TP [3174 2321]

0='SR', 1='ABQRS', 2='AFIB', 3='all_other_alone', 4='ABQRS + all_other', 5='AFIB +
all_other', 6='ABQRS + AFIB', 7='ABQRS + AFIB + all_other' 8='invalid_combination'

Training combination-class confusion matrix:
     0    1    2    3    4    5    6    7    8
0[3740  553    0  231  231  115    8  104 1852]
1[ 393  383    0   47  131   23    7   73  528]
2[   2    2    0    7    0  126    0  184    1]
3[1607  440    0  397  291  336    5  280 1113]
4[  79   76    0   36   78   25    6   62  107]
5[   0    2    0    9    5  210    0  272    1]
6[   0    3    0    3    3   25    1  106    1]
7[   0    0    0    1    3   11    1   41    1]
8[   0    0    0    0    0    0    0    0    0]

Training multi-label classification report:
              precision    recall  f1-score  support
SR             0.593518  0.817237  0.687639   6834.0
ABQRS          0.247643  0.699201  0.365746   2254.0
AFIB           0.482946  0.956905  0.641919   1021.0
all_other      0.571112  0.422384  0.485616   5495.0
micro avg      0.478432  0.670277  0.558334  15604.0
macro avg      0.473805  0.723932  0.545230  15604.0
weighted avg   0.528431  0.670277  0.567007  15604.0
samples avg    0.507094  0.659178  0.554760  15604.0
```

```
Training combination-class classification report:
                           precision    recall  f1-score       support
SR                          0.642501  0.547264  0.591071   6834.000000
ABQRS                       0.262509  0.241640  0.251643   1585.000000
AFIB                        0.000000  0.000000  0.000000    322.000000
all_other_alone             0.543092  0.088834  0.152692   4469.000000
ABQRS + all_other           0.105121  0.166311  0.128819    469.000000
AFIB + all_other            0.241102  0.420842  0.306569    499.000000
ABQRS + AFIB                0.035714  0.007042  0.011765    142.000000
ABQRS + AFIB + all_other    0.036542  0.706897  0.069492     58.000000
invalid_combination         0.000000  0.000000  0.000000      0.000000
accuracy                    0.337321  0.337321  0.337321      0.337321
macro avg                   0.207398  0.242092  0.168006  14378.000000
weighted avg                0.515427  0.337321  0.371380  14378.000000

Test metrics:
Loss:                 0.7484
Binary accuracy:      0.7003
Exact-match accuracy: 0.3288
Macro F1-score:       0.5155

Test label-wise confusion matrices:

SR:
[[501 612]
 [189 824]]

ABQRS:
[[1128  676]
 [ 130  192]]

AFIB:
[[1793  181]
 [  10  142]]

all_other:
[[1044  273]
 [ 478  331]]

Test combination-class confusion matrix:
     0   1   2   3   4   5   6   7   8
0[565  78   0  28  29  29   4  21 259]
1[ 81  39   0   7  16   5   3   9  59]
2[  1   0   0   0   2  15   0  31   2]
3[264  51   0  49  43  51   1  42 183]
4[ 14   8   0   2   8   7   1   8  10]
5[  0   1   0   1   0  26   1  27   0]
6[  0   1   0   0   1   5   3  24   0]
7[  0   0   0   1   0   1   0   9   0]
8[  0   0   0   0   0   0   0   0   0]
```

Test multi-label classification report:

	precision	recall	f1-score	support
SR	0.573816	0.813425	0.672928	1013.0
ABQRS	0.221198	0.596273	0.322689	322.0
AFIB	0.439628	0.934211	0.597895	152.0
all_other	0.548013	0.409147	0.468507	809.0
micro avg	0.460848	0.648519	0.538809	2296.0
macro avg	0.445664	0.688264	0.515505	2296.0
weighted avg	0.506388	0.648519	0.546813	2296.0
samples avg	0.488711	0.636877	0.535419	2296.0

Test combination-class classification report:

	precision	recall	f1-score	support
SR	0.610811	0.557749	0.583075	1013.000000
ABQRS	0.219101	0.178082	0.196474	219.000000
AFIB	0.000000	0.000000	0.000000	51.000000
all_other_alone	0.556818	0.071637	0.126943	684.000000
ABQRS + all_other	0.080808	0.137931	0.101911	58.000000
AFIB + all_other	0.187050	0.464286	0.266667	56.000000
ABQRS + AFIB	0.230769	0.088235	0.127660	34.000000
ABQRS + AFIB + all_other	0.052632	0.818182	0.098901	11.000000
invalid_combination	0.000000	0.000000	0.000000	0.000000
accuracy	0.328786	0.328786	0.328786	0.328786
macro avg	0.215332	0.257345	0.166848	2126.000000
weighted avg	0.503850	0.328786	0.351263	2126.000000